End-to-End Air Quality Index (AQI) Prediction
and
Real-Time Forecasting System for Lahore

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Program: 10Shine Data Science Internship Program

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Live Project Link:

https://aqi-predictor-u2yvyxwfzqtskvxkddpkv3.streamlit.app

GitHub Repository:

https://github.com/hadeebajaved/AQI-Predictor

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1. Abstract

Air pollution, particularly excessive amounts of particulate matter like PM2.5 and PM10, has been identified as one of the most crucial environmental issues in Lahore. For this project, a full-fledged MLOps pipeline has been developed to predict the value of PM2.5 in the coming 24 hours and forecast the overall trend in air pollution for the coming 3 days. With a machine learning predictive model using the Random Forest algorithm, MongoDB for storing the features and GitHub Actions for automation, users will gain real-time insights via the interactive Streamlit web app.

2. Introduction & Problem Statement

The worsening condition of air pollution in large cities needs to be tracked via an intelligent monitoring system. While weather apps can show the current condition of air pollution in your city, they fail to provide predictions based on AI and interactions among different pollutants. In this project, a complete self-sufficient machine learning system has been built that not only predicts future PM2.5 values depending upon PM10, CO, NO2, and time-based features but also does the job of retraining itself every day.

3. Architecture and Technology

To ensure high scalability and minimize manual intervention, this system transitions from a conventional static model into a dynamic, automated framework. This evolution is supported by the following core components:

Data storage (Backend): A NoSQL architecture via MongoDB is employed to manage and store all historical atmospheric datasets efficiently.

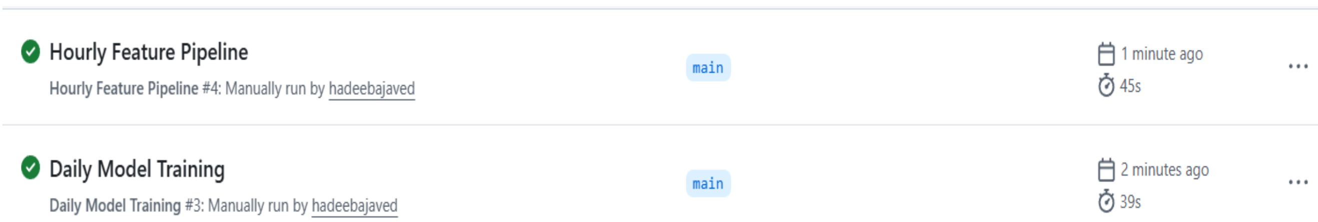
Continuous Integration/Continuous Deployment: GitHub Actions are utilized to orchestrate two primary automated workflows:

Hourly Pipeline: Retrieves live sensor data to update the database on an hourly basis.

Daily Pipeline: Facilitates the regular retraining of the AI model with fresh data every 24 hours.

Machine Learning: Predictive modeling is powered by the Scikit-Learn library, specifically utilizing the Random Forest Regressor.

Front-end Application: An interactive user interface is deployed through the Streamlit Cloud platform.

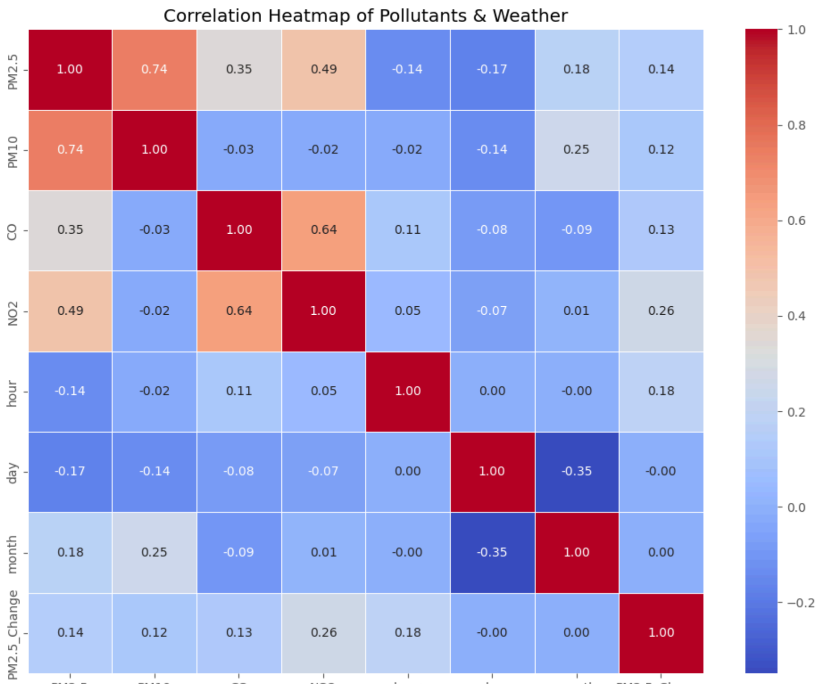


4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted before model training in order to identify the hidden trends in the atmospheric data.

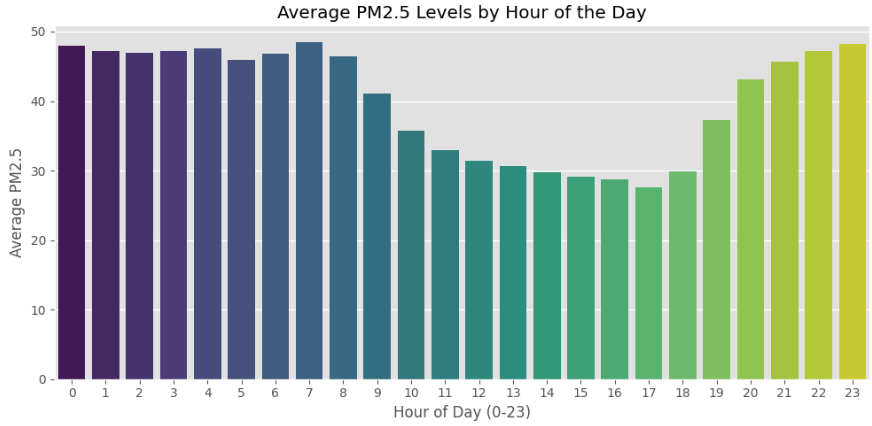
4.1. Correlation Analysis

From the correlation heatmap analysis, it was identified that there was a very high positive correlation between PM10 and PM2.5 concentration. It was also noticed that NO2 concentration also had a substantial effect on particulate matter due to vehicle emissions.

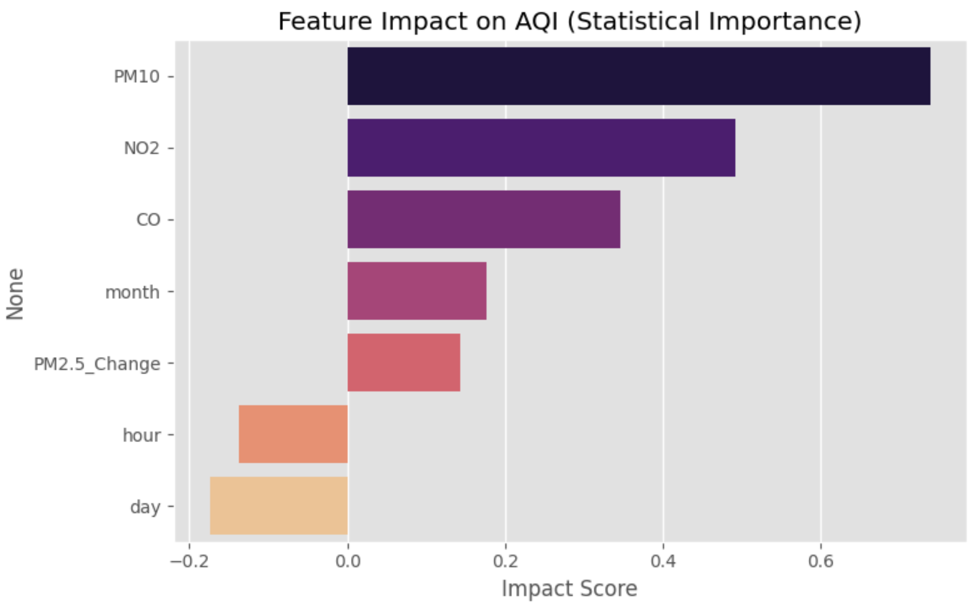


4.2. Time-Series Analysis & Hourly Trend

On conducting time-series analysis for 24 hours, different trends were discovered. PM2.5 concentration was observed to have sharp peaks during late night and early morning hours due to temperature inversion effect and heavy traffic in Lahore.



4.3. Feature Importance & Explainable AI (SHAP) Feature importance and advanced AI explainability were implemented using the **SHAP (SHapley Additive exPlanations)** library. This provides transparent insights into the model's decision-making process, indicating how historical PM2.5 concentrations, PM10 levels, and specific hours of the day drive the current predictions. This advanced analytics feature allows end-users to understand exactly which pollutants are influencing the forecast at any given hour.



5. Machine Learning Approach

Algorithm Selection: Random Forest Regressor was selected based on its good performance in predicting non-linear data and dealing with complicated relationships among multiple pollutants without overfitting.

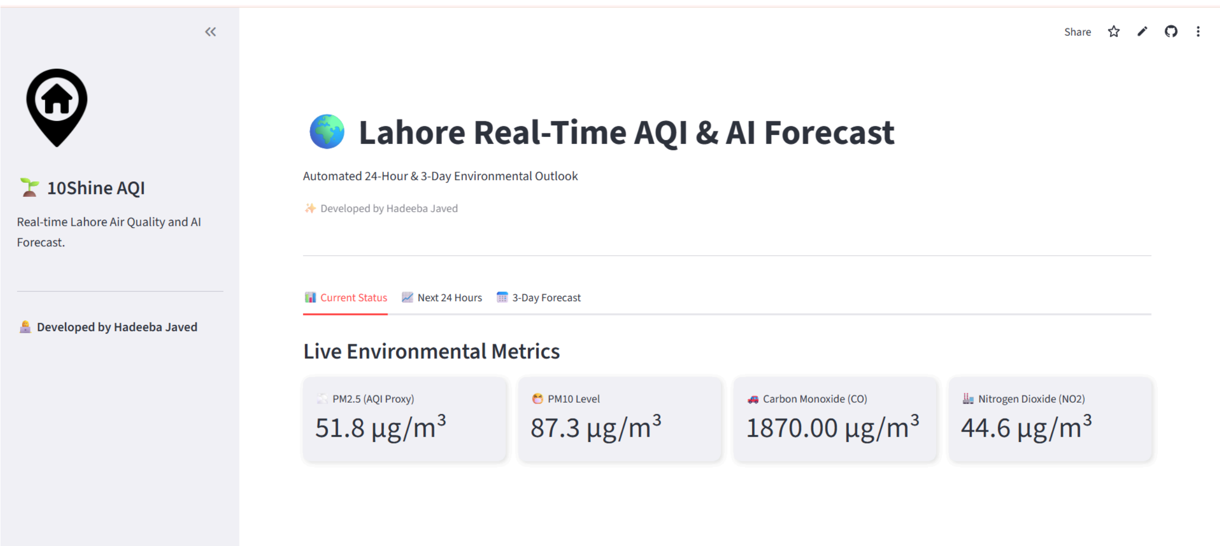
Feature Generation: Time-related features like hours, days, and months were generated. Moreover, PM2.5_Change was generated as a feature for monitoring the rate of pollution build-up.

Evaluation: Evaluation is an automated process performed during the daily execution of the MLOps pipeline. The trained model demonstrated robust forecasting capabilities on real-world data, achieving an **R^2 Score of 0.76 (76%)**, a **Mean Absolute Error (MAE) of 7.97 µg/m^3**, and a **Root Mean Squared Error (RMSE) of 11.11 µg/m^3**. These metrics confirm that the model possesses strong generalization capabilities, protecting the system against overfitting while accurately predicting Lahore's volatile environmental conditions.

6. Web Dashboard Functionality

This frontend app converts complicated machine learning predictions into useful data for all audiences.The web dashboard is composed of the following interactive modules:

- **Live Environmental Measurements:** Shows real-time values of PM2.5, PM10, CO, and NO2 measurements presented in metrics format.
- **Hazardous AQI Alert Mechanism:** A continuous monitoring system that displays a high-priority UI alert if live PM2.5 concentrations exceed the safe threshold (e.g., > 75 µg/m³), actively advising users to take health precautions.
- **Next 24 Hours Projection:** This Plotly interactive area chart represents the AI-projected hourly trend, which allows making plans for safe outdoor activities.
- **3-Day Projection:** The prediction model represented through color code (Good - Green; Moderate - Yellow; Unhealthy – Red).
- **Advanced Analytics & AI Explainer (SHAP):** A dedicated tab utilizing SHAP values to visualize feature importance for the latest data batch, allowing users to transparently see which specific pollutants are driving the current PM2.5 predictions.



7. Conclusion

This project successfully demonstrates the end-to-end lifecycle of a scalable Data Science solution, specifically designed for real-time environmental monitoring in Lahore. By transitioning from a static machine learning approach to a fully automated MLOps architecture, the system seamlessly integrates continuous data extraction, daily model retraining, and cloud deployment via GitHub Actions and Streamlit. The Random Forest predictive model proved highly robust in forecasting volatile air quality trends, achieving an impressive R^2 score of 0.76 and a remarkably low Mean Absolute Error (MAE) of 7.97 $\mu g/m^3$. Furthermore, the integration of SHAP for advanced AI explainability and a dynamic alert system for hazardous PM2.5 levels ensures that the dashboard is not only accurate but also transparent and user-centric. Ultimately, this automated data pipeline provides a reliable, maintenance-free tool that empowers citizens to make informed decisions regarding their health and daily outdoor activities.